DEVOPS CAPSTONE PROJECT

CloudDesk

Enterprise IT Helpdesk & DevSecOps Platform

GitHub Actions | Trivy | Terraform | AWS EKS | Helm | ArgoCD | Prometheus | Grafana

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github.com/bharath-kadali/devsecops-project

CloudDesk: Enterprise IT Helpdesk & DevSecOps Platform

Problem Statement

- Modern IT organizations need scalable, resilient and secure platforms to manage infrastructure incidents, access grants and system anomalies.
- The goal: do it without manual overhead.

Solution

- CloudDesk is a full-stack, cloud-native IT Helpdesk platform.
- Automated quality gates and security scanning
- Infrastructure as Code
- Declarative Kubernetes orchestration
- Comprehensive observability

DevSecOps Mission

- Shift security left
- Automate validation in CI
- Manage infrastructure declaratively with Terraform
- Achieve continuous GitOps delivery with ArgoCD on AWS EKS

In one line: every commit is tested, scanned, containerized and deployed to Kubernetes through an automated, auditable pipeline.

End-to-End Modern Cloud-Native Stack

Application Layer

- **Frontend:** React 18, Vite 6, modern CSS with dark/light themes
- **Backend:** FastAPI 0.115.6, Uvicorn, SQLAlchemy 2.0, Python 3.12
- **Persistence:** PostgreSQL 16 Alpine with Alembic migrations

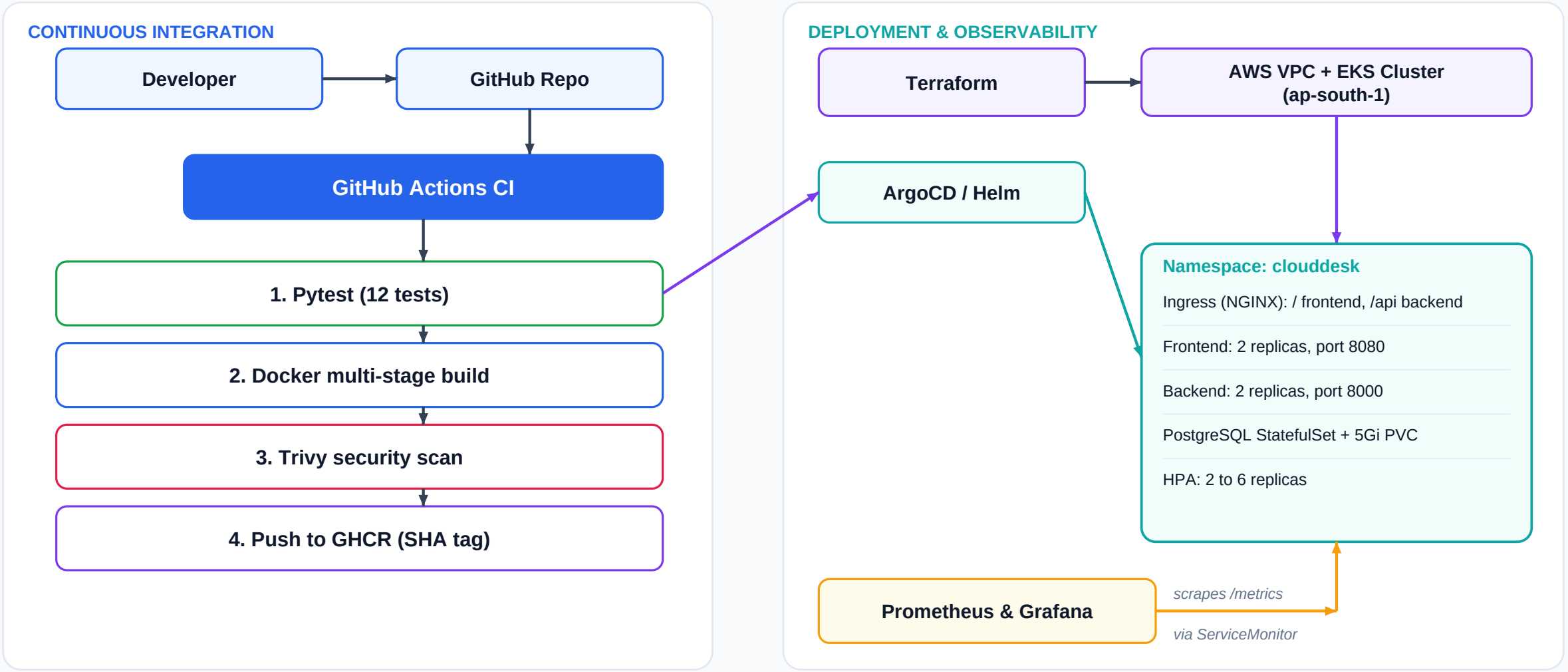
Quality & Security Layer

- **Testing:** Pytest with 12 unit/integration tests and a SQLite test DB
- **Container Security:** Aqua Security Trivy, blocking on HIGH/CRITICAL findings

Infrastructure & Platform

- **Containers:** Docker multi-stage builds (non-root `appuser`, `nginx-unprivileged`)
- **CI/CD:** GitHub Actions with GHCR
- **IaC:** Terraform 1.7+ (AWS VPC and EKS modules)
- **Orchestration:** Kubernetes 1.31, Helm 3
- **GitOps:** ArgoCD continuous reconciliation
- **Observability:** Prometheus, Grafana, ServiceMonitor

End-to-End DevSecOps Architecture



Shifting Security Left with GitHub Actions & Trivy

Pull requests and pushes to `main` trigger `.github/workflows/ci-cd.yml`

Job 1: Test

- Pytest runs 12 tests against a SQLite test database
- Build fails immediately on any failing test
- Frontend validated via `npm ci && npm run build`

Job 2: Build

- Docker images built for backend and frontend
- Immutable Git commit SHA tags:
- `clouddesk-backend:<sha>`
- `clouddesk-frontend:<sha>`

Security Gate

- Trivy CLI scans both container images
- `--exit-code 1 --severity HIGH,CRITICAL --ignore-unfixed`
- Any finding fails the pipeline

Registry Push

- Only clean, verified images are published
- Destination: GitHub Container Registry (GHCR)

Why SHA tags matter: they prevent configuration drift and the mutable-deployment risks of the `:latest` tag, so every running workload maps back to an exact Git commit.

Reproducible Cloud Infrastructure on AWS

Modular Design

terraform-aws-modules/vpc/aws

2 public + 2 private subnets across `ap-south-1a` and `ap-south-1b`; single NAT gateway

terraform-aws-modules/eks/aws

EKS 1.31 cluster with a managed node group of `t3.medium` instances (min 2, max 4, desired 2)

Quality & Safety

- HCL formatted with `terraform fmt`
- `terraform validate` passes without warnings
- `terraform.tfvars.example` provided: safe configuration without committing credentials
- Full lifecycle: `terraform init` through a clean `terraform destroy`

2	2 + 2	1.31	2 to 4
Availability Zones	Public / Private Subnets	EKS Version	Worker Nodes

Cloud-Native Release Management with Helm

Helm chart: `helm/clouddesk`

Parameterized Values

`values.yaml` enables environment-specific configuration.

High Availability

Minimum 2 replicas each for frontend and backend.

Non-Root Pod Security

Backend runs as UID 10001; frontend as UID 101 on port 8080.

Ingress Routing

NGINX Ingress sends `/api` to backend (8000) and `/` to frontend (80).

Autoscaling

Horizontal Pod Autoscaler targets 60% CPU utilization.

Smoke Testing

Ephemeral Kind cluster validation in the CI pipeline.

Full-Stack Visibility with Prometheus & Grafana

Metrics Instrumentation

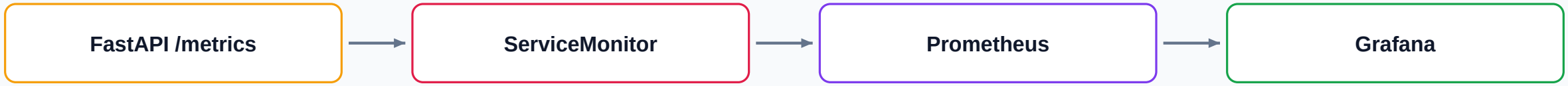
- `prometheus-fastapi-instrumentator` exposes native OpenMetrics on `/metrics`
- Tracks request count, duration, error rates and system resource consumption

Prometheus Operator

- Custom **ServiceMonitor** automatically discovers backend endpoints every 15 seconds
- Targets page verifies `clouddesk-backend` is in **UP** state

Grafana Visualization

- Real-time dashboard
- HTTP throughput: `http_requests_total`
- p95/p99 latency: `http_request_duration_seconds`
- Pod CPU and memory utilization



Capstone Reflection & Next Steps

Key Learnings

- Designing security into container images from day one (unprivileged users, minimal base images) drastically reduces vulnerability remediation overhead.
- Immutable SHA image tagging combined with GitOps ensures absolute reproducibility between Git commits and running cluster workloads.
- Separating configuration into Helm values and Terraform variables keeps clear boundaries between infrastructure and application lifecycle.

Future Enhancements

- Migrate in-cluster PostgreSQL to **AWS RDS Aurora Serverless** for multi-AZ disaster recovery.
- Integrate **HashiCorp Vault / AWS Secrets Manager** via External Secrets Operator.
- Add synthetic load generation and automated canary rollouts via **Argo Rollouts**.